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Field Application of Nitrogen-Polymer Foam Fracturing in Low-Pressure Reservoirs: Pilot Results from Arystanovskoye Field

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Abstract

The application of foam-based hydraulic fracturing fluids in low-pressure, mature reservoirs has gained attention globally as a means to minimize formation damage and optimize proppant placement. In 2024, a pioneering fluid frac operation using a nitrogen foam system was conducted in the CIS region for the first time. The operation was executed on two offset wells with similar geological and petrophysical properties. One well was stimulated using a conventional linear gel system, while the other utilized a nitrogen-foam-based fracturing fluid. Both treatments were accompanied by comprehensive microseismic monitoring and tracer diagnostics.

Microseismic analysis revealed that the foam-based treatment led to more controlled fracture height growth and longer effective fracture half-lengths, indicating a more targeted stimulation of the pay zone. Production data further demonstrated superior post-frac performance from the foam-treated well, with a 1.8x increase in initial oil rate compared to the conventionally treated offset. This paper presents an in-depth engineering evaluation of the treatments, supported by field data, simulation results, and diagnostic interpretations. The results confirm the potential of nitrogen foam fracturing for enhancing recovery in depleted, water-prone formations and set the stage for broader deployment of this technology across mature oil fields in the region.

Introduction

Hydraulic fracturing remains one of the most effective stimulation techniques for enhancing oil recovery in tight or depleted formations. Traditional gel-based fracturing systems, while effective in many scenarios, can present significant challenges in mature reservoirs with low pressure and high water cut. These include excessive water usage, risk of formation damage, poor cleanup, and vertical fracture growth into water-bearing layers.

To address these challenges, foam-based fracturing systems have been increasingly adopted in North America and the Middle East. Foam systems—particularly those based on nitrogen—offer several operational and reservoir advantages, including lower fluid leakoff, minimized clay swelling, improved

proppant transport in narrow fracture widths, and enhanced clean-up through gas expansion during flowback.

Despite growing international adoption, the technology had not been applied in the CIS region until mid-2024, when a pilot project was executed as a collaboration between KNOC, Alash Energy Services and PeWeTe Kazakhstan. The project involved two wells — Well A-434 and Well A-240—completed in the similar productive formation, Jurassic horizon. Well A-434 underwent a conventional gel-based treatment, while Well A-240 was treated with a nitrogen-foam-based fluid system.

A distinguishing feature of this project was the simultaneous use of microseismic monitoring, a first for foam fracturing operations in the region. This allowed for unprecedented insights into fracture propagation behavior under different fluid types.

This paper documents the planning, execution, and evaluation of the first foam-proppant fracture stimulation in the CIS. Detailed operational parameters, microseismic interpretations, and post-treatment production results are presented, along with a comparison to historical performance using conventional fluids. The findings contribute to the growing body of global evidence supporting the use of foam fluids in specific geological and operational contexts, especially in depleted and water-sensitive reservoirs.

Global Development and Applications of Foam Fracturing

Foam-based fracturing fluids have a long development history, starting from the early 1980s when their advantages for minimizing fluid leakoff and improving cleanup efficiency were first recognized ([Grundmann and Lord, 1983](#); [Mack and Baumgartner, 1986](#)). Since then, technology has been progressively deployed across different regions and reservoir types. In North America, foam fluids primarily nitrogen-based were successfully applied in unconventional shale such as the Eagle Ford, where reduced liquid volumes and enhanced proppant transport provided significant operational benefits ([Nath and Xiao, 2017](#)). Similar studies in other tight and shale formations also demonstrated the ability of foam to achieve improved fracture conductivity, reduced water saturation, and more effective post-frac cleanup compared to conventional crosslinked gels. In the Middle East, foamed treatments have been trialed in tight carbonate reservoirs, highlighting their ability to limit liquid invasion and provide better fracture containment under harsh downhole conditions.

In the CIS region, the earliest mentions of foam fracturing were exploratory cases in Western Siberia ([Oussoltsev et al., 2008](#)), yet systematic adoption remained absent until the present pilot. The global evidence consistently shows that foamed fracturing fluids are most advantageous in reservoirs characterized by low pressure, high water cut, or strong sensitivity to water-based gels. Recent advances—such as microseismic monitoring and non-radioactive tracer diagnostics—have enabled better quantification of fracture geometry and proppant placement under foam treatments, providing a stronger technical basis for broader field deployment.

This study therefore contributes not only as the first documented application in the CIS region, but also as part of the broader global development trend, demonstrating that foam-based fracturing is moving from niche pilot projects toward wider acceptance in mature oilfields worldwide.

Field Description

The Arystanovskoe field is located in the southern part of Western Kazakhstan and represents a mature oil asset currently in a late stage of development. The field is characterized by a combination of geologic complexity, low reservoir pressure, and variable reservoir quality, which collectively present challenges to conventional stimulation techniques.

The primary target of development and stimulation activities is Jurassic formation, a low-permeability siliciclastic reservoir with heterogeneous lithology. Core and log analyses reveal that the formation consists predominantly of fine- to very-fine-grained sandstones interbedded with siltstones and clay-rich

layers. Mineralogical composition includes significant volumes of kaolinite and illite, forming pore-filling aggregates, which increase sensitivity to fluid invasion and clay swelling. Quartz grains often exhibit secondary overgrowths, further reducing effective porosity.

Porosity across the formation varies between 12-17%, while permeability ranges from 1 to 2 mD. These values, coupled with low initial reservoir pressures (~60 bar in the case of Well A-240), classify the reservoir as tight and partially depleted.

Historical production data from Arystanovskoe demonstrate declining oil rates and rising water cut, reflecting progressive depletion and increasing communication with water-bearing zones. Previous hydraulic fracturing attempts using high-viscosity gels achieved only moderate success. These conventional treatments often resulted in unconfined fracture height growth, water breakthrough, and limited post-fracture production uplift, especially in zones with initial high water cuts.

Correlation analysis performed on prior stimulation campaigns (Fig. 1) revealed several important trends:

- Higher pumping rates lower the post frac oil rate due to fracture height is breaking to watered horizons or frac is connecting to higher water saturation zone.
- Higher the initial water cut, lower the post frac oil rate. Thus, conventional frac design cannot be pumped in initial high water cut horizons as it will increase post frac w/c and reduce oil rate.
- Higher the proppant concentration lower the post frac oil rate (-0.34), usually higher prop conc lead to higher packing of fracture width but during frac it also increases frac pressure that may lead to breaking in height into water horizon.
- Higher pad volumes and proppant concentrations correlated negatively with post-frac oil rate, likely due to induced communication with watered zones.
- Wells with higher initial oil rates showed better post-frac performance, indicating the importance of reservoir potential in determining frac success.

To address these challenges, a novel foam-based fracturing approach was proposed for Well A-240, targeting the same 13-meter pay interval within the Jurassic horizon as the offset Well A-434, which received a conventional gel-based treatment. The pilot was designed to evaluate the effectiveness of nitrogen-foam-based fluids in depleted, water-sensitive formations, using microseismic monitoring and post-job tracer diagnostics to understand fracture geometry, containment, and fluid transport efficiency.

This comparative setup between the two wells offers a unique opportunity to isolate the effects of fluid system choice on stimulation outcomes under similar reservoir and operational conditions.

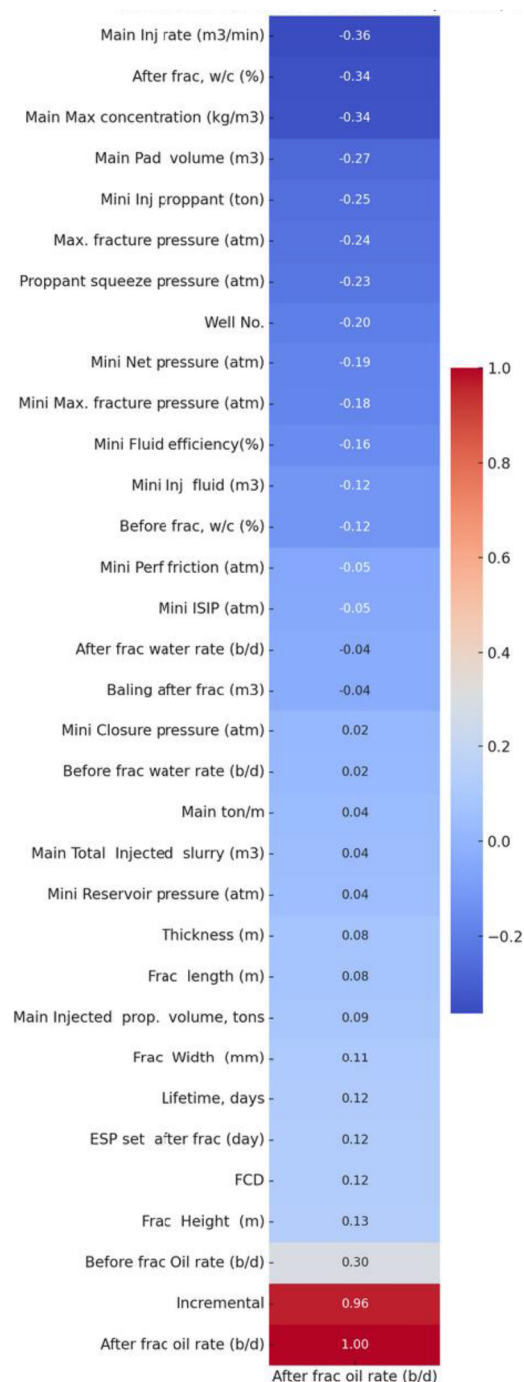


Figure 1—Correlation Matrix of Post-Fracturing Oil Rate with Treatment and Reservoir Parameters.

Treatment Design

The stimulation design for both wells —A-434 (conventional) and A-240 (foam-based)—was developed to address the key challenges associated with the Jurassic formation: low pressure, high water cut, limited pay thickness and presence of water-sensitive minerals. The primary objective was to achieve effective fracture half-length extension and minimizing the risk of out-of-zone growth.

Design Constraints and Objectives

Previous stimulation campaigns in the field showed that high pad volumes and excessive proppant mass tended to correlate with elevated post-frac water cut and poor oil rate performance. Therefore, the design strategy emphasized:

- Controlled fracture height growth
- Decreased pad and proppant volumes
- Improved clean-up efficiency

The perforated interval in both wells was about 13 meters, selected based on geological correlation and production history. The same zone was targeted to ensure comparability of post-fracture response.

Conventional Fracturing Design (Well A-434)

To minimize the risk of excessive water production in the high-permeability zones of the J-VIIB reservoir (Fig. 2), a focused perforation strategy was implemented. The interval 2,707–2,719 m was specifically selected to promote fracture height containment within a safe range, approximately up to 2,727 m. While this interval minimizes vertical fracture growth into adjacent water-bearing layers, some uncertainty remains due to the absence of a distinct stress barrier within the J-VIIB formation.

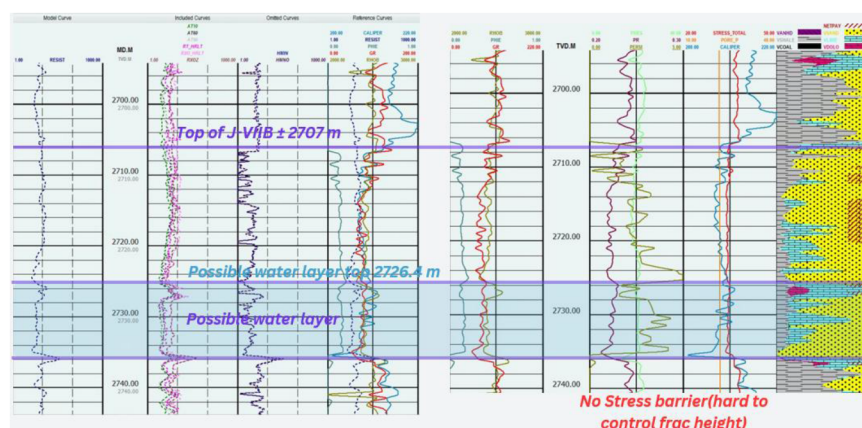


Figure 2—Geomechanical Cross-Section of Well A-434.

The selected perforation zone is characterized by favorable reservoir properties, including high effective porosity, low minimum horizontal stress, and the lowest recorded Young's modulus and Poisson's ratio within the target section. These mechanical attributes indicate reduced rock stiffness and limited lateral strain response. Caliper logs and mechanical property interpretation, which predicted a fracture azimuth aligned with the prevailing stress regime (Fig. 3). Verification of fracture orientation was planned by using microseismic mapping during the hydraulic fracturing operation.

Conservative pumping schedule was designed to minimize out-of-zone growth and reduce fluid exposure to water-sensitive intervals. The job included 15 metric tons of proppant with a 30# gel loading, approximately 90 m³ of fluid, and a pumping rate of 3.6 m³/min. This contrasts with a previous-campaign conventional proppant fracturing design, which might involve a more aggressive approach — for example, a 45# gel, 50 tons of proppant, and 170 m³ of total fluid volume.

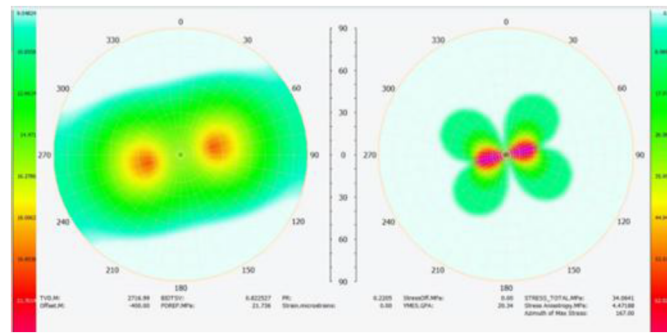


Figure 3—Predicted Fracture Azimuth.

Fracture height was estimated at approximately 22 meters, with a tendency for fracture growth to shift toward the lower part of the perforated interval. This behavior correlates with the lower minimum horizontal stress and higher permeability in that zone.

The effective fracture half-length was estimated at approximately 40 meters and was concentrated in the interval with the highest conductivity. This optimized fracture geometry ensured effective stimulation of the most productive portion of the reservoir, thereby maximizing reservoir contact and production potential within the target zone.

Foam-Based Fracturing Design (Well A-240)

Well A-240, targeting the Jurassic J-XI interval (Fig. 4), was identified as a candidate for advanced stimulation due to its significantly depleted reservoir pressure and low historical oil production. The bottomhole pressure (BHP) of A-240 was estimated at approximately 28 bar, one of the lowest in the field. Despite this, the cumulative oil recovery from the well remained minimal. Comparatively, other wells such as A-108 and A-223—positioned structurally higher and near faulted zones—demonstrated sustained and higher production performance, with measured pressures of ~84 bar and ~76 bar, respectively, as of 2021.

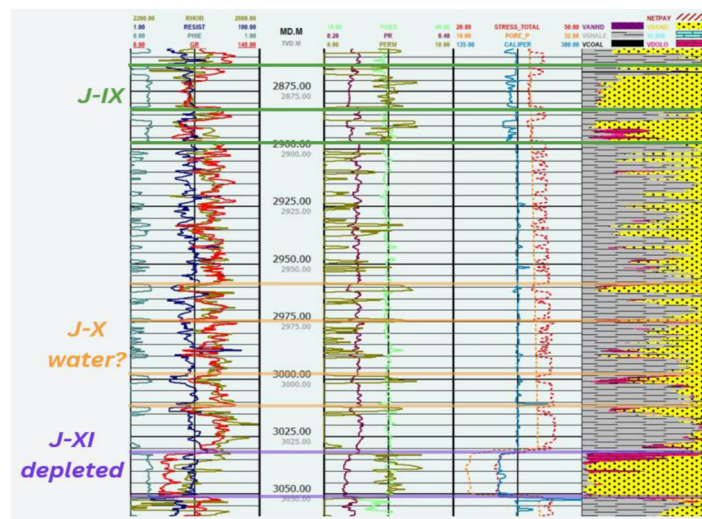


Figure 4—Geomechanical Cross-Section of Well A-240.

Pre-fracture oil production from A-240 was around 0.5 m³/day, emphasizing the need for stimulation. Geological evaluation revealed that the highest permeability and production potential were likely associated with the J-IX formation, while the J-XI interval appeared to be more depleted. Additionally, the interbedded, clay-rich laminated sandstones of the J-X subzones were interpreted as water-bearing with relatively low permeability.

Based on log analysis and mechanical properties, the re-perforation interval of 3,033–3,046 m was selected with deep penetration charges. Fracture azimuth prediction based on caliper log and mechanical property interpretation indicated a potential alignment parallel to the nearby fault. Final fracture orientation and geometry were planned to be confirmed through microseismic monitoring during the stimulation.

A tailored nitrogen-foam fracturing design was developed, marking the first application of foamed proppant fracturing in a depleted reservoir in Central Asia. The proposed schedule included multiple proppant mesh sizes—20/40, 16/20, and 12/18—to maximize fracture conductivity. The total design volume consisted of 88 m³ of liquid, approximately 20 metric tons of proppant, and 8,000 m³ of nitrogen, ensuring effective proppant transport and cleanup under low-pressure reservoir conditions.

This operation represented a technically ambitious pilot, combining advanced diagnostics and stimulation design to evaluate the viability of foam fracturing in significantly depleted formations.

Execution Overview: Well A-240 (J-XI Interval)

The hydraulic fracturing treatment of Well A-240, targeting the Jurassic J-XI formation, was carried out in October, 2024. The operation was executed in three stages: Injection test, MiniFrac and MainFrac.

Injection Test Interpretation

The treatment began with wellbore fill and injection test (Fig. 5). During the first four minutes, fluid was pumped to fill the wellbore, and surface pressure remained at 0 atm. As pumping continued, the surface pressure increased and peaked at 410 atm at a maximum rate of 3.2 m³/min. The Instantaneous Shut-In Pressure (ISIP) was recorded at 105 atm, and surface pressure rapidly declined to 0 psi, indicating closure pressure below hydrostatics. This further highlighted the importance of using downhole gauges for pressure analysis in depleted reservoirs.

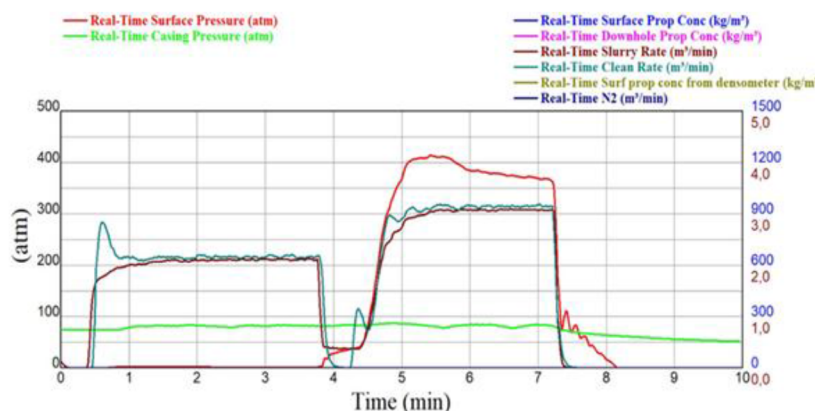


Figure 5—Injection Stage Plot for Well A-240.

The downhole pressure data proved indispensable, revealing a closure pressure of 200 atm and a high net pressure of 203 atm, attributed to reservoir depletion. This finding suggested that injection pressures above 90 atm at the wellhead could induce fracturing—critical information for future water injection planning, as excessive pressure could unintentionally initiate fractures.

MiniFrac Interpretation

Fluid was again pumped to fill the wellbore, and surface pressure remained at 0 atm for the initial six minutes. Surface pressure peaked at 403 atm with a pumping rate of 3.6 m³/min (Fig. 6). The ISIP was recorded at 130 atm, and surface pressure again dropped to vacuum levels within 30 minutes.

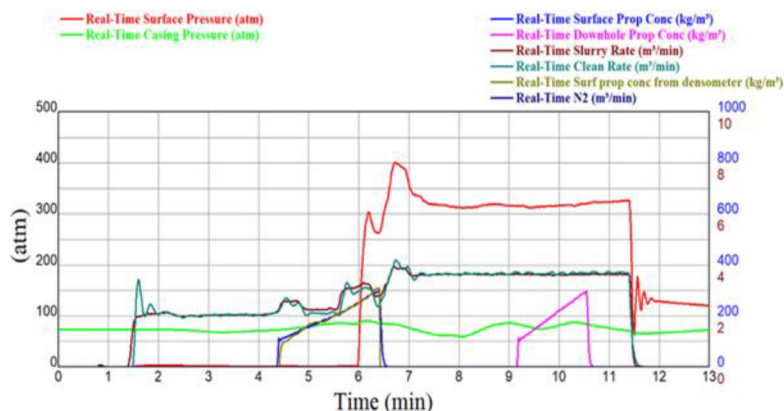


Figure 6—Minifrac Stage Plot for Well A-240.

The maximum downhole pressure during the MiniFrac reached 439 atm. Although no clear breakdown pressure was observed in the downhole plots, pipe friction was estimated at 109 atm. Fracture closure was detected approximately 15 minutes after the pressure drop began, providing a calibration point for fracture modeling and pad volume optimization.

Main Frac Operation and Results

The main stage of the nitrogen-foam-based fracturing operation proceeded according to design. A total of 130 m³ of fluid, 8,000 m³ of nitrogen (to generate ~70% quality foam), and 19 metric tons of proppant (mesh sizes: 20/40, 16/20, 12/18) were pumped. The maximum surface wellhead pressure reached 365 atm, while the downhole pressure peaked at 486 atm. Bottomhole temperature was observed to cool from 105°C to approximately 40°C during the treatment.

Post-job Simulator interpretation showed an asymmetric fracture geometry (Fig. 7): the northeast wing extended approximately 13 meters longer than the southwest, likely due to reservoir depletion on the southeast side of the well. The total fracture length was estimated at 104 meters with a fracture height of 14 meters. However, the conductive portion—the segment contributing to fluid flow—was approximately 74 meters in length and 14 meters in height. A separate fracture event cluster suggested a secondary fracture lobe extending ~120 meters laterally and 40 meters vertically to the southwest.

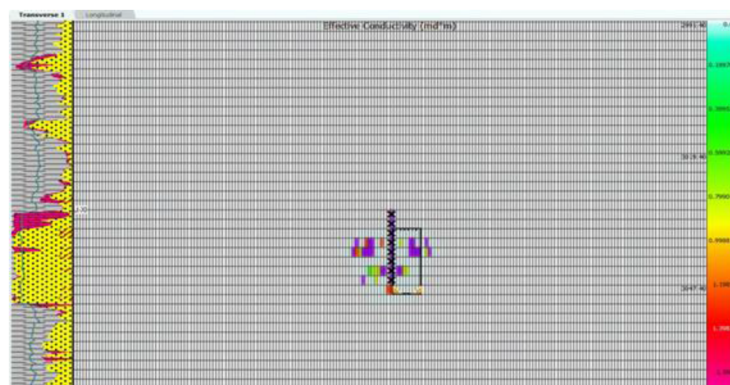


Figure 7—Calibrated Fracture Geometry Based on Pressure Matching - Well A-240.

Microseismic Monitoring

A full set of microseismic events was recorded during the stimulation. The fracture azimuth was interpreted to be approximately 43° north (Fig. 8), parallel to the nearby faults in the north and south of Well A-240,

indicating a normal faulting stress regime. The fracture initiated near the middle of the perforated interval (3,033-3,046 m), with most events recorded between 3,030 and 3,055 m.

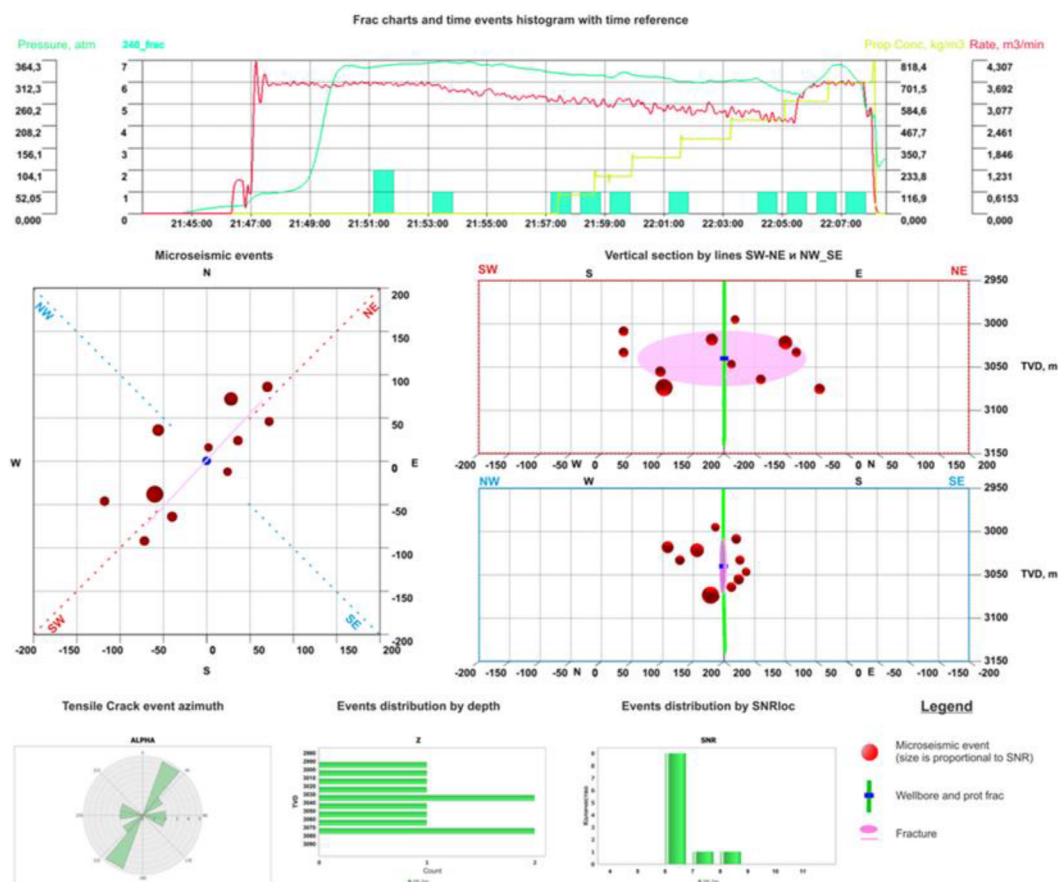


Figure 8—Microseismic Interpretation Results for Well A-240: Event Distribution, Fracture Geometry, and Azimuth.

Microseismic data showed a dominant NW-SE fracture orientation, aligned with the regional stress field, favorable for optimal reservoir drainage. The clustering of events near the wellbore suggested relatively simple fracture geometry with limited branching—ideal for effective proppant placement and sustainable production. The event size (as inferred from signal-to-noise ratios) and distribution further indicated that fracture growth remained largely confined to the target interval, minimizing the risk of out-of-zone propagation.

Tracer Diagnostics

Post-frac Non-Radioactive Tracer (NRT) proppant log interpretation confirmed proppant coverage was restricted to the reservoir section, particularly between depths of 3,031 and 3,049 m—an 18-meter interval (Fig. 9). This aligned well with microseismic data and pressure matching results, indicating effective fracture placement within the target zone.

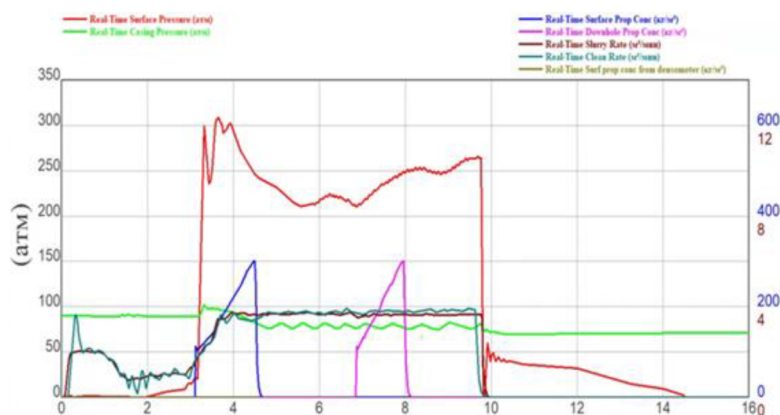


Figure 10—Minifrac Stage Plot for Well A-434.

Fracture closure occurred about 20 minutes after the liquid level began to drop (Fig. 11), providing a reliable reference point for modeling and pad volume optimization. The downhole pressure gauges also enabled after-closure diagnostic analysis. Linear flow behavior was observed between 0.349 and 0.750 hours, with bottomhole pressure declining from 23.18 MPa to 17.94 MPa, reflecting efficient fracture conductivity and strong reservoir performance. Radial flow analysis revealed a transition time of 137.35 minutes, with permeability calculated at 9.51 mD and a radial transmissibility (Kh) of 85.6 md·m. The resulting Kh/g value of 28.32 md·m/cp indicated strong reservoir flow potential under current depletion.

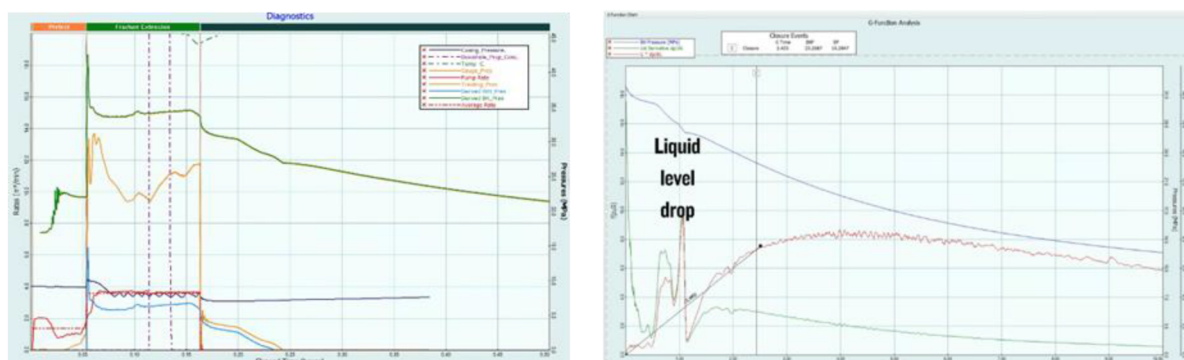


Figure 11—Minifrac Stage Analyses for Well A-434.

Main Frac Operation and Results

Following the minifrac interpretation and pressure anomaly review, the main frac was executed per original design with slight modifications to prevent screen-out risks. The total treatment included 15 metric tons of proppant: 4T of 20/40 mesh WellProp, 5T of 16/30 mesh WellProp, and 6T of 16/20 mesh RCP. Surface pressure peaked at 322 atm, while downhole pressure remained below model predictions. The bottomhole temperature dropped from 115°C to 35°C, confirming efficient fluid cooling during treatment.

Fracture geometry, interpreted through pressure-matching, showed an asymmetric pattern with the northeast wing longer than the southwest. The total fracture length was ~104 m with a height of 12 m. However, the conductive portion was ~68 m in length and 12 m in height (Fig. 12).

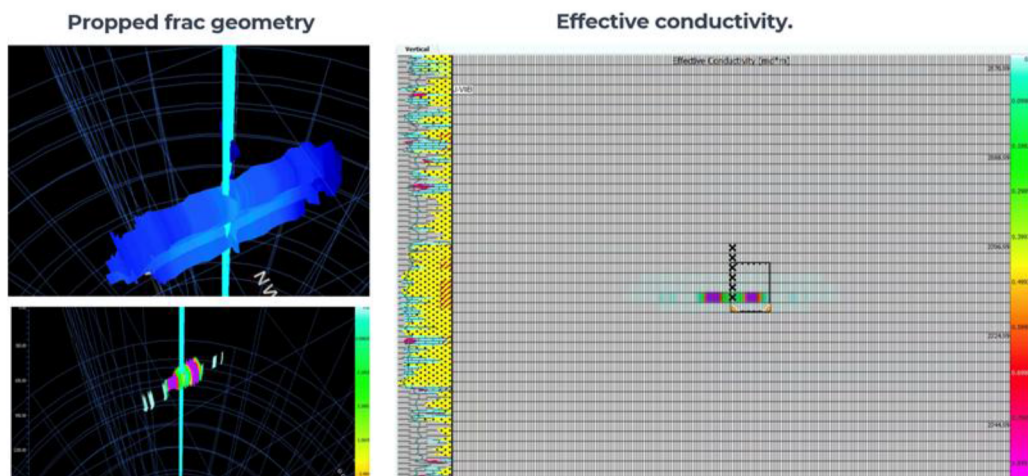


Figure 12—Calibrated Fracture Geometry Based on Pressure Matching - Well A-434.

Microseismic Monitoring

Microseismic events largely followed a SW-NE trend, with evidence of minor complexity likely induced by natural fractures or lithological heterogeneities (Fig. 13). Most events were concentrated near the wellbore, indicating controlled fracture propagation. However, a portion of the events extended outside the target zone.

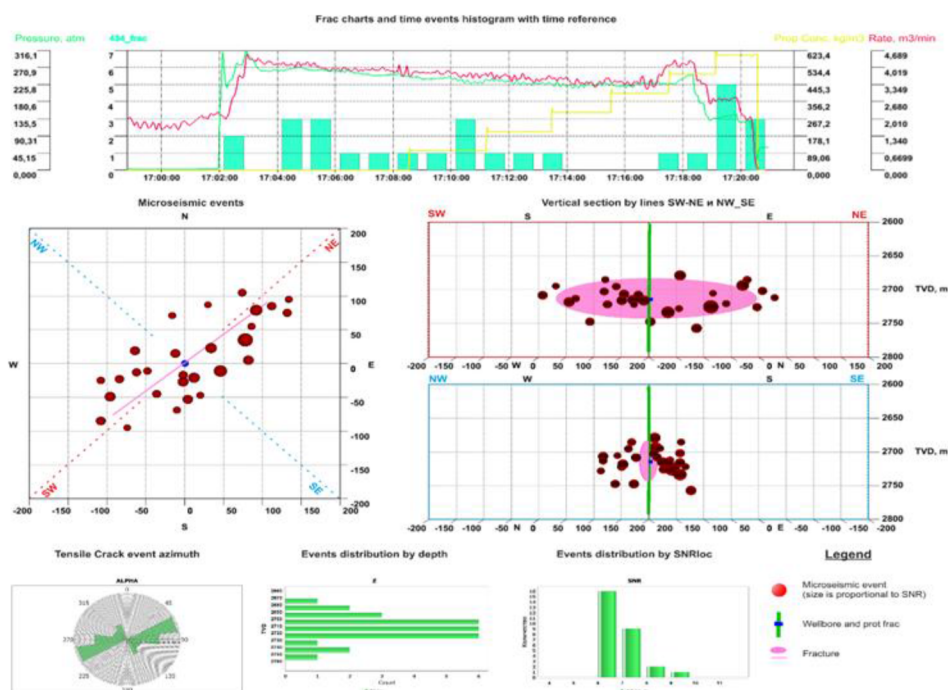


Figure 13—Microseismic Interpretation Results for Well A-434: Event Distribution, Fracture Geometry, and Azimuth.

Engineering Evaluation and Comparative Analysis

Stimulated Reservoir Volume (SRV) Comparison

A comprehensive analysis of microseismic and post-fracture data reveals a significant contrast in SRV geometry between the foam-based treatment in Well A-240 and the conventional gel-based treatment in Well A-434 (Fig. 14).

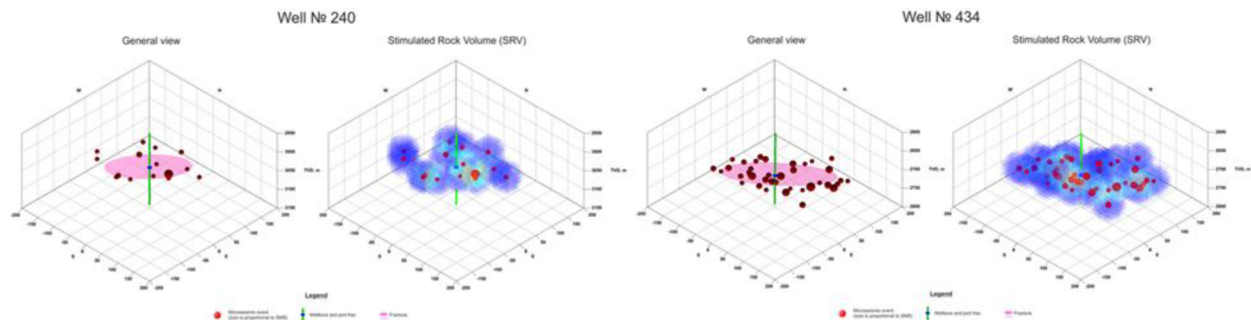


Figure 14—SRV Visualization and Fracture Geometry Comparison for Wells A-240 and A-434.

Well A-434 (Conventional Proppant Frac)

- The fracture network exhibited wider spatial distribution, both horizontally and vertically, with indications of extensive lateral propagation.
- Microseismic events extended further from the wellbore, suggesting secondary fracture development and unconfined growth, likely due to the combination of high pad volumes and elevated treatment pressure.
- This larger SRV may enhance recovery from a broader drainage area but also introduces risks of water blocks and fracture penetration into water-bearing formations.

Well A-240 (Foamed Proppant Frac)

- In contrast, the foam-based fracturing treatment created a more compact and vertically contained SRV, closely aligned with the pay zone.
- The limited fluid leak-off into the formation and smaller fracture footprint indicate efficient fluid containment and reduced damage, which is particularly beneficial in depleted and water-sensitive intervals.
- Improved clean-up efficiency was also observed, facilitated by nitrogen expansion and reduced post-job fluid handling requirements.

A more detailed comparison of the fracture geometry parameters derived from microseismic monitoring, pressure matching, and tracer diagnostics is summarized in [Table 1](#).

Table 1—Comparative Fracture Geometry Metrics for Foam- and Gel-Based Treatments.

Parameter	Well A-240 (Foam), 19 tons of proppant	Well A-434 (Gel), 15 tons of proppant
Total fracture length (m)	~104	~104
Conductive fracture length (m)	~74	~68
Fracture height (m)	~14	~12
Fracture azimuth (°)	~43	~50

Quantitative comparison of the monitored parameters shows that the nitrogen-foam frac in Well A-240 created a slightly larger fracture (14 m height and 74 m conductive length) compared to Well A-434 (12 m and 68 m, respectively). This difference is partly explained by the higher proppant loading in the foam treatment (18 metric tons vs. 15 metric tons in the gel frac). Importantly, despite the larger dimensions, microseismic monitoring and tracer diagnostics confirmed that the foam-based stimulation produced a compact, vertically contained geometry aligned with the pay zone, while the gel treatment resulted in broader and less controlled lateral propagation. These findings demonstrate that foam fracturing can provide

both effective proppant placement and superior fracture containment, even under higher proppant loads, thereby reducing the risk of out-of-zone growth and unwanted water production. This contrast underscores the importance of aligning fracturing fluid systems with reservoir properties and stimulation objectives.

While the conventional frac approach maximized SRV and achieved a higher treatment volume, it posed a greater risk of uncontrolled fracture growth and fluid migration, particularly in pressure-depleted settings.

On the other hand, the foam-based frac proved highly effective in delivering targeted stimulation, with minimal water production risks, enhanced conformance, and greater operational efficiency—especially critical in low-pressure, high water cut environments.

Conclusions

- Foamed fracturing (Well A-240) was effective in achieving controlled fracture propagation and zonal containment, proving suitable for depleted and water-sensitive reservoirs.
- Conventional fracturing (Well A-434) resulted in greater SRV and fracture length, which may be advantageous in tighter formations but requires careful monitoring to mitigate risks of water breakthrough and out-of-zone stimulation.
- Lower gel loadings at elevated reservoir temperatures proved operationally stable, with no screen-out events recorded, demonstrating efficient proppant transport even under challenging conditions.
- Fracture geometry was non-linear and more complex than predicted by conventional models, reaffirming the need for advanced diagnostics and real-time monitoring.
- Downhole pressure gauges are essential in depleted reservoirs to identify accurate breakdown and closure pressures.

This evaluation demonstrates that fracturing fluid selection must be tailored to both reservoir characteristics and development objectives, and that diagnostic technologies are essential to maximize stimulation effectiveness while minimizing risk.

Acknowledgments

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